

An Organisational Cybernetics-Based System for Comprehensive Academic Performance Evaluation in Higher Education

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Abstract

This paper addresses the ongoing challenge of evaluating academic performance in large, complex higher education institutions. The aim is to present a comprehensive evaluation system grounded in organisational cybernetics that fully integrates formative assessment and contextual variables. Traditional approaches tend to privilege quantitative indicators and static metrics, often neglecting the dynamic, multifaceted nature of academic work. By embedding context-aware formative assessment into the performance evaluation process, the proposed model seeks to foster adaptive learning and continual improvement.

Purpose: This paper addresses the ongoing challenge of evaluating academic performance in large, complex higher education institutions. The aim is to present a comprehensive evaluation system grounded in organisational cybernetics that fully integrates formative assessment and contextual variables. Traditional approaches tend to privilege quantitative indicators and static metrics, often neglecting the dynamic, multifaceted nature of academic work. By embedding context-aware formative assessment into the performance evaluation process, the proposed model seeks to foster adaptive learning and continual improvement.

Design/Methodology/Approach: This study employs a targeted literature review and conceptual modeling using Beer's Viable System Model (VSM) to map the academic performance evaluation process. The process is structured across the five VSM subsystems (S1–S5) and multiple recursion levels within the faculty, with formal communication and feedback channels facilitating systemic adaptation. The evaluation integrates formative assessment mechanisms through the OADI (Observe–Assess–Design–Implement) learning cycle, promoting ongoing improvement and responsiveness to institutional needs. By aligning the evaluation system with the VSM framework, the approach ensures both theoretical robustness and practical relevance for complex university environments.

Findings: The resulting framework provides a recursive and adaptive structure in which formative assessment is embedded at every organisational level, supporting alignment between individual, departmental, and institutional objectives. This structure enables evaluation processes that are both responsive to local contexts and consistent with overarching institutional strategy. The inclusion of contextual factors, reflective practice, and qualitative data supports a balanced approach that fosters both accountability and academic development.

Originality/Value: This cybernetic framework advances the field by demonstrating how organisational cybernetics can support a multidimensional, contextually responsive performance evaluation system in higher education. By systematically incorporating formative assessment and organisational learning into the evaluation process, the approach overcomes key limitations of traditional models and promotes continuous improvement, innovation, and faculty development. The framework offers valuable insights for policy, leadership, and the future design of academic evaluation systems.

1 Introduction

In recent decades, institutions of higher education worldwide have faced mounting pressure to reevaluate and improve the ways they measure academic performance among faculty members. This imperative has emerged in response to evolving expectations of quality, accountability, transparency, and continuous improvement in the university sector (OECD, 2019). Within engineering education, the complexity of faculty roles has grown significantly. Far from being limited to classroom teaching and research, contemporary academic responsibilities now include active engagement with external stakeholders, technical consulting, administrative management, and the cultivation of university-society linkages (Boyer, 1997; Cadez, Dimovski, & Zaman Groff, 2017). This multidimensionality echoes Boyer’s expansive notion of academic scholarship, which asserts that the academic missions of teaching, research, service, and engagement have equal importance in the modern university.

As a consequence, the field of academic performance evaluation has evolved rapidly from traditional, unidimensional models—often privileging research output—to more inclusive, multicriteria frameworks designed to encompass the full breadth of scholarly work (Felder & Brent, 2004; Hicks et al., 2015). Many universities now seek to integrate teaching, research, extension, and management as co-equal domains in their annual assessments (Asún & Zúñiga, 2017; OECD, 2019), not only as a recognition of the broadened nature of academic work but also as a means of aligning institutional practices with global benchmarks such as those established by ABET or CDIO (ABET, 2023). Modern evaluation systems thus tend toward the multidimensional, embracing both quantitative and qualitative assessments negotiated through weighted scoring models, peer evaluation processes, and the increasing use of digital tools for evidence collection (Felder, Brent, & Prince, 2011; Villanueva et al., 2017).

At the same time, the rise of digital platforms—such as electronic portfolios and analytics-enabled dashboards—has substantially enhanced the capacity for transparent, traceable, and evidence-based performance management in academia (Álvarez Sepúlveda, 2020; Bayas Escudero, 2022; Gonzales-Saji et al., 2021). By facilitating the centralization of data sources, the automation of evidence gathering, and the integration of diverse metrics, such tools are helping institutions move toward best practices in academic performance assessment.

Nevertheless, significant methodological and structural challenges persist in the implementation of robust evaluation systems in higher education. Among the most pervasive issues is the persistence of structural biases, where research achievements often outweigh teaching and service in practical recognition and reward structures (Cadez et al., 2017; Bornmann, 2016), to the detriment of teaching-focused or engagement-intensive academics. This implicit hierarchy has the potential not only to demotivate those delivering excellence in undervalued domains but also to undermine the mission of producing “well-rounded” academic professionals (Boyer, 1997). Furthermore, the diversity of timelines and forms of evidence associated with distinct academic functions—such as long-term research projects, short-term teaching engagements, or episodic outreach initiatives—makes fair comparison inherently complex (Cipagauta-Moyano, 2019; Clavijo-Cáceres & Balaguera-Rodríguez, 2020).

Metrics customarily used across academia, including bibliometric indicators (publication and citation counts) and student evaluations, have well-established limitations. Bibliometric indices, such as the h-index, have been shown to inadequately reflect disciplinary differences and overlook teaching-focused work, while also being vulnerable to manipulation and not always aligning with the intended impact or quality of research (Aksnes, Langfeldt, & Wouters, 2019; Ding, Liu, & Kandonga, 2020). Similarly, student course evaluations—which often serve as the primary quantitative input for teaching effectiveness—are subject to various forms of bias, typically exhibit low response rates, and may fail to reliably capture true educational impact (Berk, 2005; Rizo Moreno, 1999; Santos del

Real, 2012). These methodological shortcomings raise concerns about the validity and fairness of these evaluation tools, potentially eroding trust in the system among faculty and diminishing its value as a driver for improvement (Asún & Zúñiga, 2017).

Compounding these challenges are issues related to the qualitative assessment of functions such as community engagement, extension, or academic leadership. Evaluating such domains is complicated by the inherently narrative and context-dependent nature of the evidence available, which often resists standardized scaling and introduces subjectivity into committee deliberations (Felder & Brent, 2004; OECD, 2019). Without formalized criteria and calibration mechanisms, peer and committee evaluations can become inconsistent or even arbitrary, further exacerbating faculty concerns about fairness and comparability (Espejo, 2007; Espejo & Reyes, 2011). This, in turn, may disincentivize faculty participation in domains deemed inadequately valued or unfairly assessed, running counter to contemporary calls for balanced academic missions.

Against this backdrop, the need for a systemic, integrative, and adaptive approach to academic performance evaluation becomes manifest. Recent literature advocates for drawing on the principles of organizational cybernetics—particularly Stafford Beer’s Viable System Model (VSM)—as a robust conceptual foundation for addressing the organizational complexity inherent in higher education (Beer, 1979; Espejo, 1990; Espejo, 1991). Under the cybernetic paradigm, a viable organization is defined not solely by its current functionality, but by its capacity to survive, adapt, and thrive in evolving environments through self-regulation and organizational learning (Beer, 1979; Espejo & Harnden, 1989). Beer’s VSM asserts that organizations are recursively structured: universities comprise faculties, faculties comprise departments, and departments comprise individuals or groups, with each level operating autonomously within its own environment, yet connected through recursive governance and feedback loops (Beer, 1979; Espejo & Reyes, 2011).

Prior applications of VSM to higher education have demonstrated that explicit recursive modeling of management subsystems at each organizational level is crucial for distributing autonomy, fostering adaptability, and ensuring cybernetic viability (Ashford, 1993; Espejo & Reyes, 2011; Rezk et al., 2019). The literature further highlights the importance of communication channels and feedback mechanisms in organizational learning, both to propagate high-level goals downward into actionable criteria and to feed upward real-world evidence for policy refinement (Beer, 1985; Espejo, 2013; Hicks et al., 2015).

The theoretical synergies between organizational cybernetics and contemporary models of academic evaluation are further reinforced by emerging best practices in higher education. The literature on performance assessment emphasizes the imperative to employ multiple, triangulated sources of evidence for each academic domain (Felder & Brent, 2004; OECD, 2019), recognizing that no singular metric is either sufficiently comprehensive or unbiased. For teaching, research, extension, and management, a balanced combination of portfolios, direct outcomes, peer assessments, and standardized metrics has emerged as the gold standard (Molero López-Barajas & Ruiz Carrascosa, 2005; Villanueva et al., 2017; Felder et al., 2011).

In the domain of teaching evaluation, the literature is notably critical of excessive reliance on student feedback surveys, which—though valuable as one data point—should be complemented by peer observations, curated teaching portfolios, and outcomes-based measures (Berk, 2005; Felder & Brent, 2004; Arbesú García & Gutiérrez Martínez, 2014). Teaching portfolios, comprising syllabi, instructional materials, innovation reports, and reflective narratives, are increasingly viewed as crucial instruments for capturing the full spectrum of pedagogical practice (Bozu, 2011; Galán-Mañas, 2019). Their growing adoption is linked to improvements in reflective practice, professional growth, and the institutional recognition of broader contributions to student learning (Jerez Yáñez et al., 2016; Álvarez Sepúlveda, 2020; Bayas Escudero, 2022).

For research evaluation, there has been a strong policy turn toward the “responsible use of metrics”

(School of Biological Sciences & School of Clinical Medicine, University of Cambridge, 2021; DORA, 2024), with guidelines such as the Leiden Manifesto (Hicks et al., 2015) urging that indicators be interpreted in context, triangulated with qualitative judgment, and adapted to disciplinary diversity. The recognition of research outputs beyond peer-reviewed articles—including patents, software, industry projects, and evidence of societal impact—reflects an expanded understanding of what constitutes meaningful scholarly contribution, particularly within engineering and applied science contexts (Bornmann, 2016; Lebel & McLean, 2018; Prancut , 2021; Mart nez-Garrido & Murillo, 2024). Additionally, best practice now dictates adjusting expectations according to academic load, career stage, and research profile, thus avoiding one-size-fits-all standards which may unfairly penalize teaching- or management-focused faculty (Fritch et al., 2021; Hicks et al., 2015; Tao et al., 2025).

Traditional, linear performance appraisal systems are insufficient for the diverse and often conflicting demands of faculty roles in engineering, which include teaching, research, management, and external engagement (OECD, 2019; Cadez et al., 2017). Static, quantitative evaluations cannot capture the “variety” of activities, temporalities, and evidence types. The complexity of higher education—involving diverse stakeholders, autonomy, and rapid change—necessitates more adaptive approaches (Espejo & Reyes, 2011).

Systemic and cybernetic frameworks, such as Beer’s Viable System Model (VSM), provide tools for managing this complexity through feedback and learning mechanisms (Beer, 1979; Espejo, 1990). VSM structures organizations into five recursive subsystems (S1–S5), each requiring feedback, adaptability, and alignment (Beer, 1985). Applying the VSM can clarify roles, reduce bureaucracy, and balance autonomy with control, following Ashby’s Law of Requisite Variety (Ashby, 1956).

Best practices in evaluation incorporate triangulated indicators (teaching, research, engagement, management) and both quantitative and qualitative methods (Felder & Brent, 2004; Hicks et al., 2015). Portfolios, peer evaluation, and standardized rubrics enhance transparency and comparability ( lvarez Sep lveda, 2020). Integrated digital platforms and robust data stewardship have become essential due to the increasing volume and heterogeneity of evaluative data (Bayas Escudero, 2022; Gonzales-Saji et al., 2021).

Key challenges persist: standardization, data governance, qualitative assessment, bias, and resistance to change (OECD, 2019; As n & Z niga, 2017; Hatch & Curry, 2020). International frameworks now emphasize multidimensional, context-responsive, and continually refined evaluation systems, integrating feedback and learning cycles for institutional excellence (DORA, 2024; Argyris, 1977; Espejo, 2013; Espejo, 2014).

2 Methodology

2.1 Systems-Cybernetic Framework

This study applies the principles of organizational cybernetics and systems thinking, with a particular emphasis on Stafford Beer’s Viable System Model (VSM), to inform the design and analysis of an academic performance evaluation system (Beer, 1979; Espejo & Harnden, 1989). The VSM provides a recursive, prescriptive template for mapping the necessary and sufficient subsystems that enable an organization—such as an engineering faculty—to remain viable and adaptive within a dynamic environment (Beer, 1985; Espejo & Reyes, 2011).

2.2 Functional Decomposition Using the Viable System Model

The research begins by systematically modeling the faculty as a constellation of nested viable systems, each comprising five core subsystems:

- Operational units (e.g., teaching, research, engagement, management), responsible for the creation of organizational value
- Coordination mechanisms to maintain stability and minimize conflict among operational areas
- Control and resource allocation functions, ensuring performance optimization and organizational homeostasis
- Intelligence and planning subsystems that scan the external environment and foster strategic adaptation
- Policy and identity functions that set purpose and maintain cohesion (Beer, 1985; Espejo & Harnden, 1989)

This structure is explicitly aligned at multiple organizational levels (individual, department, faculty) in keeping with the recursive logic central to the VSM (Espejo & Reyes, 2011).

2.3 Managing Complexity and Variety

A core tenet of the methodological approach is the management of organizational variety, operationalized through Ashby’s Law of Requisite Variety—namely, that only variety can absorb variety (Ashby, 1956; Beer, 1979). The methodology addresses the inherent diversity of faculty roles and evaluation evidence by employing a mix of regulatory mechanisms:

- Reducers (norms, rubrics, unified scoring scales) standardize processes and attenuate excessive complexity
- Amplifiers (information technology, flexible agreements, digital portfolios) expand the system’s responsiveness to local needs (Beer, 1985; Espejo & Reyes, 2011)

This balance is continuously monitored to maintain both local autonomy for departments and cohesion at the faculty level.

2.4 Coordination, Control, and Audit Channels

Effective coordination is facilitated by standardized protocols and shared evaluation calendars, supporting procedural fairness and reducing interdepartmental conflict (Espejo, 1990). Distributed control is administered via dedicated committees responsible for performance management and strategic planning, while an independent audit function strengthens accountability by validating data integrity and evaluation outcomes (Beer, 1985; Espejo & Reyes, 2011).

2.5 Organizational Learning and Continuous Improvement

Organizational learning is embedded in the evaluation system through both single-loop (incremental improvement) and double-loop (revision of goals and norms) feedback processes (Argyris, 1977; Espejo, 2013). The methodology operationalizes a cyclical process of Observe–Assess–Design–Implement (OADI), enabling rapid adaptation and adjustment in response to internal performance trends and external demands (Kim, 1993). Committee-based review and regular recalibration of criteria and instruments are central features.

2.6 Data Governance and Information Infrastructure

The sustainability and integrity of the evaluation system rely on robust digital infrastructure and explicit data governance protocols (Bayas Escudero, 2022; OECD, 2019). Data stewardship, confidentiality, and audit trails are emphasized to ensure that the volume and diversity of evidence can be managed transparently and securely (Gonzales-Saji et al., 2021).

2.7 Ethical Safeguards and Participative Calibration

The approach prioritizes balance between autonomy and organizational control, drawing on Beer’s axiom that vertical controls should not stifle necessary local variety (Beer, 1979). Regular participation of academic staff in reviewing procedures and criteria, along with independent mechanisms for appeal and review, mitigate the risks of bias and foster trust in the process (Cadez et al., 2017; OECD, 2019).

3 Results

The process of academic performance evaluation constitutes a complex organizational system that seeks to balance autonomy, control, and learning within the university structure. From Stafford Beer’s perspective (1979, 1985), the viability of such a process relies on maintaining a dynamic equilibrium between the operational independence of academic departments and the institutional coherence of the faculty as a whole. This equilibrium enables the organization to adapt to environmental changes without losing its core identity or purpose (Beer, 1985; Espejo & Reyes, 2011).

The Viable System Model (VSM) provides a framework for representing and understanding this structure through five interrelated systems (S1–S5). Within this paradigm, each recursion level—faculty and department—is conceived as a viable system in its own right. At each level, System 1 comprises the fundamental operations that sustain the existence and purpose of the system: at the faculty level, these are the departments; at the departmental level, they are the individual and collective academic activities that constitute academic performance (Beer, 1985).

3.1 Recursion Level N: Faculty

System 1 – Operations (Academic Departments)

(See Figure 1: Faculty – Recursion Level N)

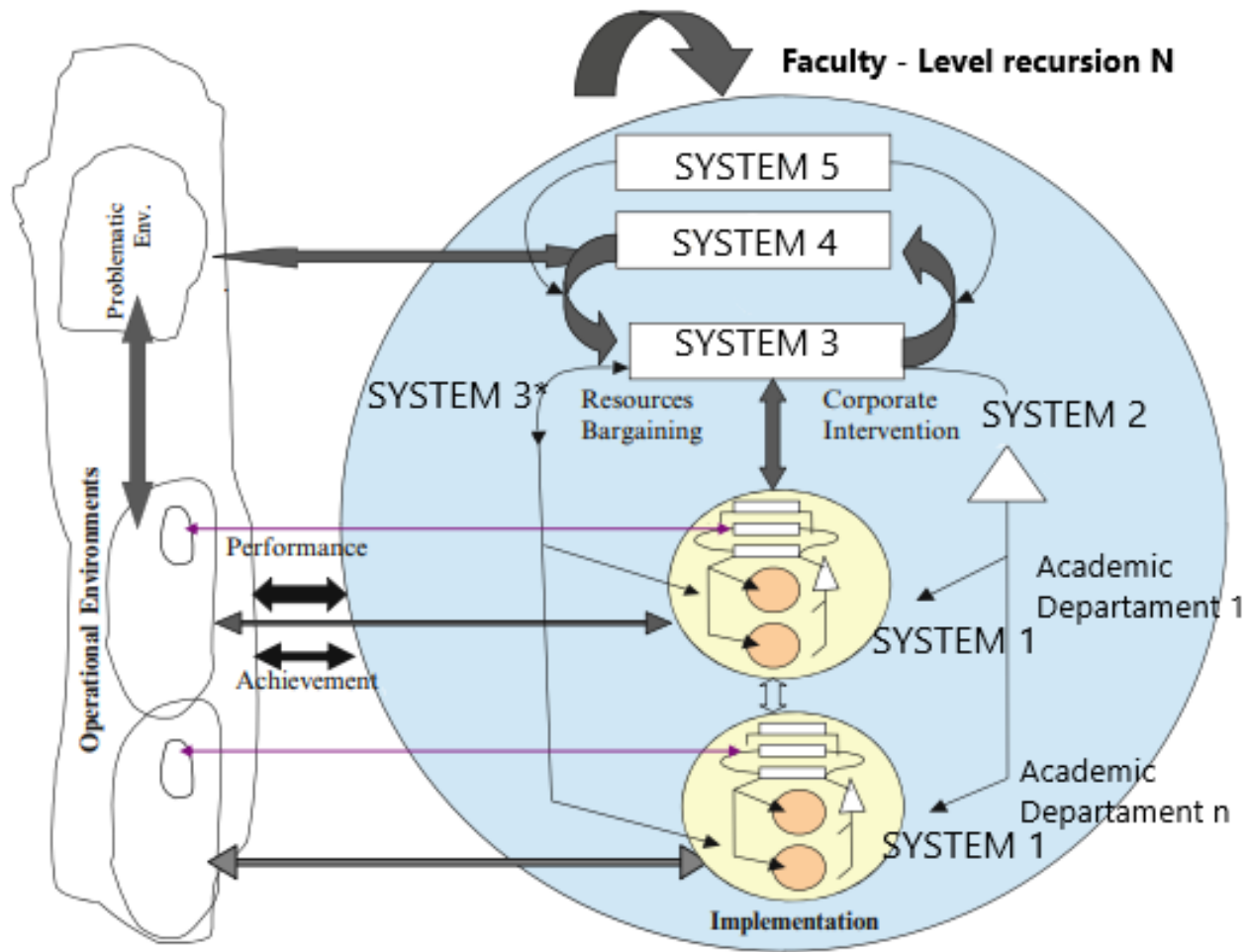


Figure 1: Faculty – Recursion Level N

Source: Espejo and Reyes (2011).

At the faculty level, academic departments serve as the System 1 units—the operational entities responsible for carrying out the core functions of the higher-level system. Each department implements the processes of assessment and performance review as defined by institutional regulations (e.g., Resolution 5949/2009), which include the planning, execution, and evaluation of academic commitments in five institutional dimensions:

- Teaching
- Research
- Extension and Community Engagement
- Technical Assistance
- Academic Management

From a cybernetic standpoint, each department operates with functional autonomy within boundaries set by the faculty as the higher system. Departments define their own internal mechanisms

for monitoring, evidence review, and decision-making; however, they maintain structured connections with the meta-level (deanship and evaluation committees) to report results, receive guidelines, and adjust their actions to align with broader institutional objectives (Beer, 1979; Espejo & Harnden, 1989).

Each System 1 unit establishes its own annual evaluation cycle, which encompasses three main phases:

- Planning: Declaration of academic commitments and annual goals
- Execution: Implementation of activities and documentation of evidence
- Evaluation: Analysis of achievement and determination of outcomes

These processes are communicated to System 3 (deanship and faculty management) through formal channels (such as reports, meeting minutes, and institutional records) as well as informal channels (such as peer feedback and interdepartmental coordination).

In summary, at the faculty level:

- System 1 units are the academic departments, embodying the operational variety of the institution.
- Each department contributes essential information to the processes of regulation and organizational learning.
- Departments preserve disciplinary and methodological autonomy within a shared normative framework.

3.2 Recursion Level N–1: Academic Department

System 1 – Operations (Individual and Collective Academic Activities)

(See Figure 2: Academic Department – Recursion Level N–1)

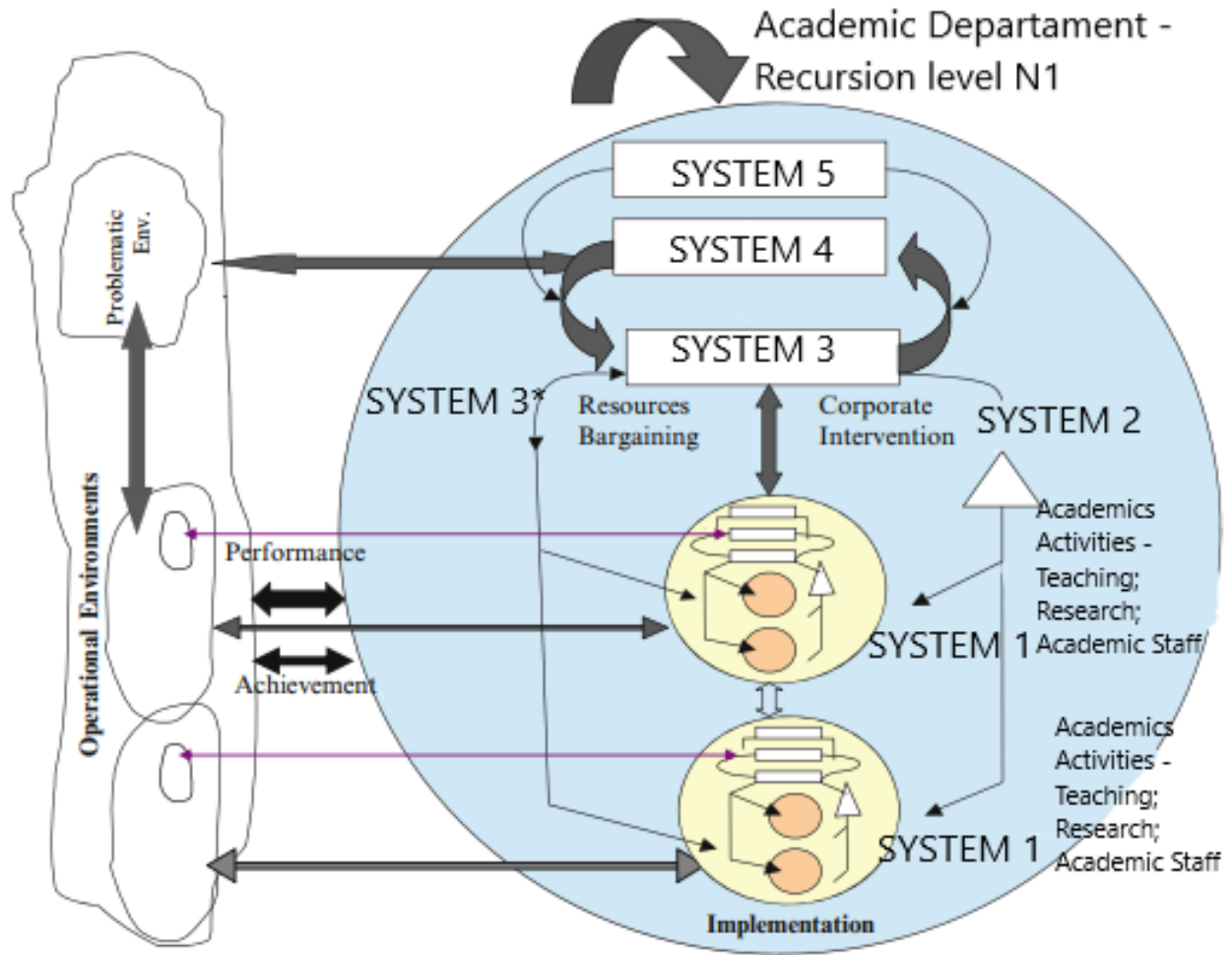


Figure 2: Academic Department – Recursion Level N-1

Source: Espejo and Reyes (2011).

At recursion level N-1, each academic department itself functions as a viable system, with its System 1 corresponding to the academic activities carried out by individual faculty members and work groups across the five performance dimensions.

Each faculty member, work stream, or academic subunit represents an operative subsystem that produces verifiable outputs: courses taught, research projects, publications, outreach initiatives, thesis supervision, or program management. Together, these outputs comprise the operational core of the department and determine its capability to generate academic value (Beer, 1979; Espejo & Reyes, 2011).

The performance evaluation process at this level is structured around:

- Individual submission of academic commitments and corresponding evidence
- Review by the Departmental Evaluation Committee, which acts as the local regulatory body
- Consolidation of results to be communicated to the higher-level (faculty) system

Each System 1 unit at the department level (i.e., the work of each academic or group) maintains professional autonomy but is connected to the system through a shared normative framework,

ensuring coherence and traceability. In cybernetic terms, the variety generated by individual practices is absorbed by departmental coordination and control mechanisms before reaching the faculty level, thus helping to preserve the stability and viability of the entire system (Beer, 1985; Espejo & Reyes, 2011).

Thus:

- The department is constituted as a viable system with its own System 1 units (individual academics and work groups).
- Each academic operates as a viable system at a lower recursion level, regulated by shared evaluation criteria.
- The faculty consolidates the results from all these systems into a coordinated and coherent institutional regulatory process.

3.3 Stage 1. Declaration of Academic Commitments

Process: Performance Agreement between Department Chair and Faculty Member

(See Figure 3: System in Focus – Academic Department Level)

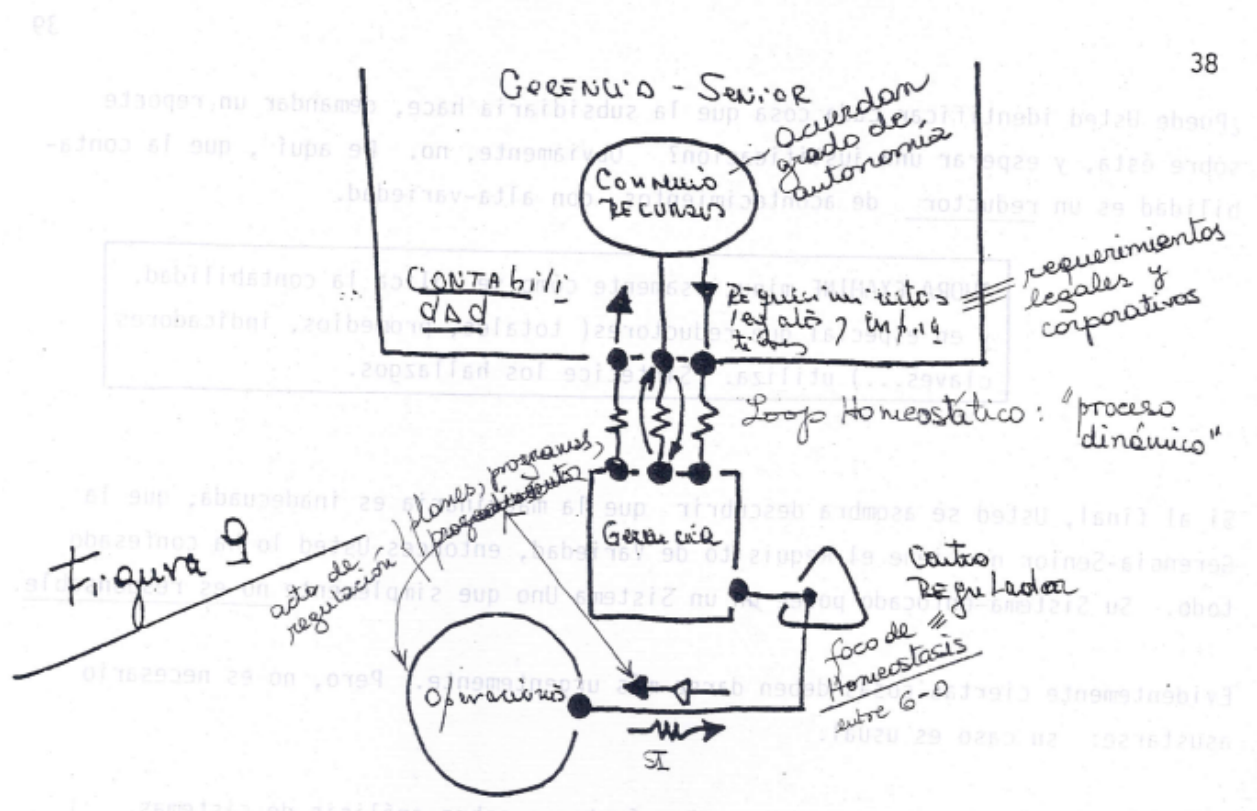


Figure 3: System in Focus – Academic Department Level

Source: Beer (1985).

1. General Description of the System in Focus In the first stage of the academic performance evaluation process, the system in focus is situated at the recursion level of the academic department, specifically within the interaction between the Department Chair (local management and regulatory subsystem) and the faculty member (operational subsystem).

This process consists of the formal declaration of annual academic commitments, documented through a Performance Agreement. This agreement establishes weekly and semester teaching loads, as well as the targets, indicators, and evidence that will guide subsequent evaluation.

According to institutional policy (e.g., Resolution 5949/2009), this stage marks the onset of the evaluation cycle and embodies a negotiation process between department leadership and the academic staff. The process is grounded in principles of equity, autonomy, and alignment with institutional objectives (Beer, 1985; Espejo & Reyes, 2011).

The agreement is managed via an institutional digital platform, which acts as a parameterized and traceable system, ensuring transparency and effective control. This platform distinguishes three distinct user profiles:

- Faculty Profile (S1): Records commitments, workloads, and evidence.
- Chair Profile (S3): Validates and adjusts targets according to departmental priorities.
- Evaluation Committee Profile (S2): Reviews, coordinates, and validates overall process coherence prior to closure.

This digital architecture mirrors the structure of the Viable System Model: System 2 ensures coordination and standardization of all System 1 (faculty) processes within the department (Beer, 1985; Espejo & Harnden, 1989).

2. Cybernetic Interpretation of the Process (Based on Figure 3)

In terms of the Viable System Model, this process represents a homeostatic coupling between two subsystems operating with distinct yet complementary logics:

- The faculty member embodies System 1 (S1), serving as the operational unit responsible for essential functions—teaching, research, outreach, technical assistance, and academic management.
- The Department Chair represents Systems 3 (local operational control) and 5 (local identity), charged with ensuring individual objectives contribute to collective departmental goals.

The figure depicts a regulatory and dynamic negotiation loop. Here, the Performance Agreement serves as a cybernetic instrument for alignment and control.

a. Homeostatic Loop (Dynamic Equilibrium) The main loop linking the faculty member (S1) and the Chair (S3) establishes a homeostatic process of continuous adjustment.

- The faculty member brings operational variety (proposed activities, interests, projects, and capabilities).
- The Chair brings regulatory variety (institutional requirements, departmental priorities, available resources).

Equilibrium—or homeostasis—is achieved when both varieties are mutually adjusted, producing a realistic set of commitments aligned with departmental and faculty objectives (Beer, 1979; Espejo & Reyes, 2011).

Within this framework, the digital platform becomes the operative feedback channel that sustains dynamic equilibrium. Its capacity to record, compare, and validate information in real time enables the system to monitor its own stability and anticipate potential misalignments.

b. Negotiation Process and Autonomy At the higher recursion level, the “Senior Manager” (Faculty) grants degrees of autonomy to the departmental level. Within this delegated framework, the Department Chair negotiates with each faculty member under the confines set by relevant policy and strategic objectives.

The digital, parameterized system functions as an environment of controlled autonomy: each actor operates within boundaries established by the Faculty. Academics may propose workloads and commitments, but the platform enforces institutional limits, preventing overallocation and misalignment with broader planning.

c. Variety Control and Transparency The subsystems responsible for accounting and recording (linked to Systems 2 and 3) ensure traceability and standardization. The digital platform:

- Automatically logs all decisions and validations,
- Enables audits and historical reviews,
- Ensures interdepartmental coherence.

These digital mechanisms operationalize System 2 (coordination), preventing oscillations between faculty members (System 1 units) and ensuring the department maintains a shared pattern of evaluative management (Beer, 1985; Espejo & Harnden, 1989).

3. The Performance Agreement as a Systemic Regulator From a cybernetic perspective, the Performance Agreement functions as a closure mechanism for the homeostatic loop by translating negotiation into an explicit set of regulated variables. These variables—targets, products, evidence, and assigned hours—reduce the complexity of the operational system (S1), making it observable, measurable, and governable by the regulatory system (S3) (Espejo & Reyes, 2011).

The agreement not only sets targets, but also establishes a shared language between operational and regulatory subsystems, enabling subsequent processes (monitoring, evaluation, feedback) to occur on a stable basis. The digital and traceable nature of the platform amplifies this role, making performance agreements verifiable audit objects and reinforcing institutional feedback and systemic learning.

4. Position within the Annual Evaluation Cycle (Resolution 5949/2009) According to institutional policy, this stage corresponds to the initiation of the annual evaluation cycle, formalizing the agreement between the parties. The outcome—the Performance Agreement—serves as the reference point for all subsequent stages:

- Monitoring and support,
- Results evaluation,
- Feedback and cycle closure.

From a VSM perspective, this initial moment establishes the set of variables that will enable future regulatory action, performing an act of variety engineering: defining autonomy boundaries, workloads, and goals that will guide evaluation through the rest of the cycle (Beer, 1985; Espejo & Harnden, 1989).

The digital system also preserves informational viability by ensuring decision traceability, transparency of criteria, and coherence across recursion levels.

5. Synthesis The process of declaring academic commitments through the Performance Agreement constitutes a first-order homeostatic mechanism for departmental evaluation. Its function is not only administrative, but essentially cybernetic: to balance autonomy and control, align individual and institutional objectives, and establish the conditions for system viability within the annual evaluation cycle.

The implementation of a parameterized, traceable digital platform transforms this process into an intelligent regulatory system, where each profile (faculty, chair, committee) acts as an interconnected subsystem, maintaining the stability, transparency, and continuous learning of the evaluation model.

3.4 Stage 2. System 2 – Coordination and Homeostatic Stability

Process: Standardization, Protocols, and Digital Evaluation Platform

(See Figure 4: System 2 of the Viable System Model – Academic Department Level)

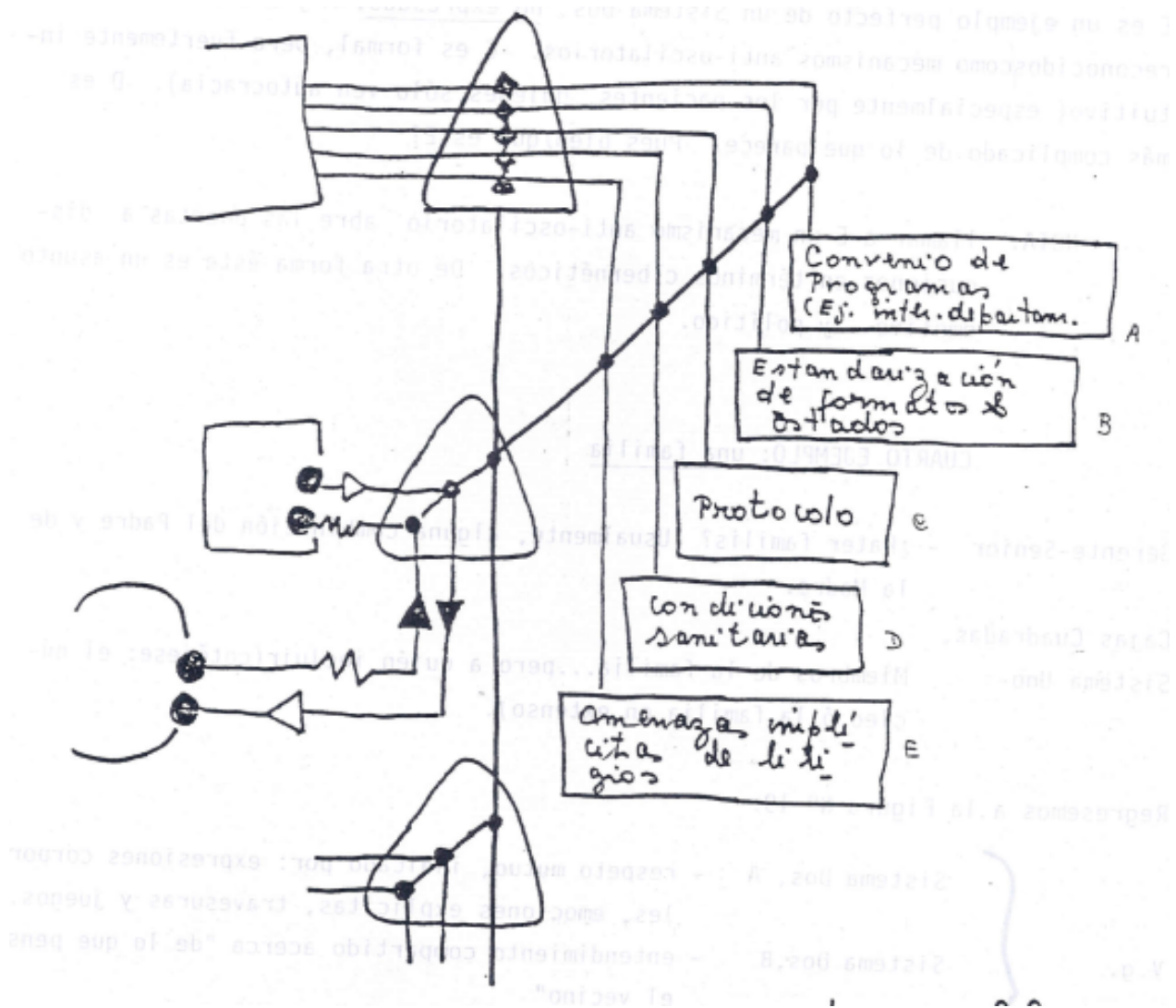


Figure 4: System 2 of the Viable System Model – Academic Department Level

Source: Beer (1985).

1. General Description of the System in Focus System 2 serves as the coordination mechanism within the Viable System Model (VSM). In the context of academic performance evaluation, its primary function is to dampen oscillations and conflicts among the various System 1 units (faculty members) within a department, ensuring that evaluation processes are procedurally consistent, equitably interpreted, and free from interference between operational entities.

Normatively, System 2 is embodied in:

- The evaluation protocols and calendars established by the faculty,
- Standardized performance templates and rubrics adopted uniformly across all operational units,
- The Departmental Evaluation Committee, which ensures instruments and criteria are applied coherently,

- And, crucially, the institutional digital platform, which orchestrates information flows and process traceability (Beer, 1985; Espejo & Reyes, 2011).

Figure 4 visually represents this structure as a series of coordinated communication pathways, where horizontal channels operate as anti-oscillatory lines, linking operational subsystems (faculty) with regulatory actors (department chair and evaluation committee).

2. Cybernetic Interpretation of System 2 In the logic of the Viable System Model (Beer, 1979, 1985), System 2 operates as a dynamic buffer between System 1 units, preventing local fluctuations—stemming from differences in interpretation, timing, or procedural approaches—from escalating into organizational instability.

In academic evaluation, such fluctuations manifest as:

- Lack of alignment between departments in applying rubrics,
- Differences in evidence review timelines or criteria,
- Divergent interpretations of assessment indicators.

System 2 stabilizes these variations through standardized protocols and digital routines, which function as variety reducers, while also deploying coordination amplifiers that increase the system’s responsiveness to contingencies and unique cases.

a. Variety Reducers Variety reducers limit complexity and interpretive dispersion:

- Standardized performance rubrics adopted by the faculty, minimizing subjectivity,
- Institutional calendars and deadlines, synchronizing evidence review processes,
- Digital validation protocols on the evaluation platform, standardizing each phase.

These mechanisms ensure that operational variety generated by System 1 units (faculty) is absorbed in a controlled manner, preventing overload at the management level (S3) (Ashby, 1956; Beer, 1985).

b. Variety Amplifiers Variety amplifiers enhance coordination and adaptability:

- A traceable, parameterized digital platform, boosting information processing and synchronous communication,
- Departmental dashboards, enabling oversight of process progress,
- Professional development for Evaluation Committees, strengthening mediation and collegial judgment,
- Asynchronous feedback channels, allowing real-time adjustment between faculty, chair, and committee (Beer, 1985; Espejo & Harnden, 1989).

c. Transducers Transducers are mechanisms that translate information across domains (e.g., qualitative data to numeric metrics, individual evidence to aggregate reports):

- Digital forms, coding academic commitments and results into observable variables,
- Electronic rubrics, converting qualitative judgments into quantitative scores,
- Consolidated reports, automatically generated to transform individual performance data into aggregate indicators for management (Beer, 1985; Espejo & Reyes, 2011).

Ashby’s Law of Requisite Variety is fulfilled when transduction channels possess at least the requisite capacity to absorb the informative diversity of the system (Ashby, 1956; Beer, 1985).

3. Key Metrics for System 2 System 2 performance can be evaluated through cybernetic indicators of stability and coherence, covering both operational effectiveness and organizational learning:

Dimension / Key Metric / Description/Instrument

- *Stability* – Average validation time variation: Measures time consistency across departments and faculty.
- *Procedural Coherence* – % of forms validated with no remarks: Evaluates procedural protocol efficacy and standardization.
- *Variety Absorption* – Ratio of corrected/total incidents: Quantifies anti-oscillatory mechanism efficiency.
- *Transduction* – Degree of qualitative–quantitative equivalence: Reflects quality of digital rubric design.
- *Institutional Learning* – Number of protocol adjustments post-review: Indicates effective feedback in the OADI cycle.

These metrics allow monitoring of whether System 2 maintains interdepartmental homeostasis, ensuring the evaluation process remains fair, stable, and replicable (Espejo & Reyes, 2011).

4. Integration with the Digital Platform and User Profiles System 2 visibly materializes in the institutional digital evaluation platform, a distributed mechanism for digital coordination and control. This platform represents the “nervous system” of the process, interconnecting three user types:

- Faculty (S1): Registers commitments, evidence, and self-evaluations,
- Department Chair (S3): Validates loads, supervises progress, ensures coherence,
- Evaluation Committee (S2): Monitors, coordinates, and guarantees criterion uniformity.

From a cybernetic viewpoint, the platform:

- Attenuates information variety through standardized forms (reducers),
- Amplifies control capacity with dashboards and real-time traceability (amplifiers),

- Transduces heterogeneous information between formats and levels (transducers),
- And offers automatic metrics to facilitate continuous feedback between recursion levels (faculty–department–academic).

Thus, digitalization does not replace human coordination, but rather augments System 2's homeostatic capacity, allowing the faculty to absorb increasing evaluation variety without losing organizational coherence (Beer, 1985; Espejo & Harnden, 1989).

5. Synthesis System 2 forms the backbone of institutional stability within the academic performance evaluation process. In the VSM perspective, this system:

- Dampens oscillations generated by the diversity of practices and rhythms among faculty,
- Coordinates and synchronizes System 1 (faculty) actions through digital routines and protocols,
- Connects and translates information between actors and recursion levels via reliable transducers,
- And guarantees traceability and transparency via the institutional digital platform, functioning as a structural communication amplifier.

Collectively, System 2 assures process homeostasis by absorbing and stabilizing operational variability before escalation to the management level (S3). In this way, System 2 performs not only a technical coordination function, but also an epistemological one: ensuring that knowledge produced by evaluation is reliable, comparable, and useful for organizational learning (Ashby, 1956; Beer, 1985; Espejo & Reyes, 2011).

3.5 Stage 3. System 3* – Independent Audit and Verification

Process: Monitoring and Validation Mechanism between Faculty and Departments

(See Figure 5: System 3* of the Viable System Model – Faculty Recursion over Departments)

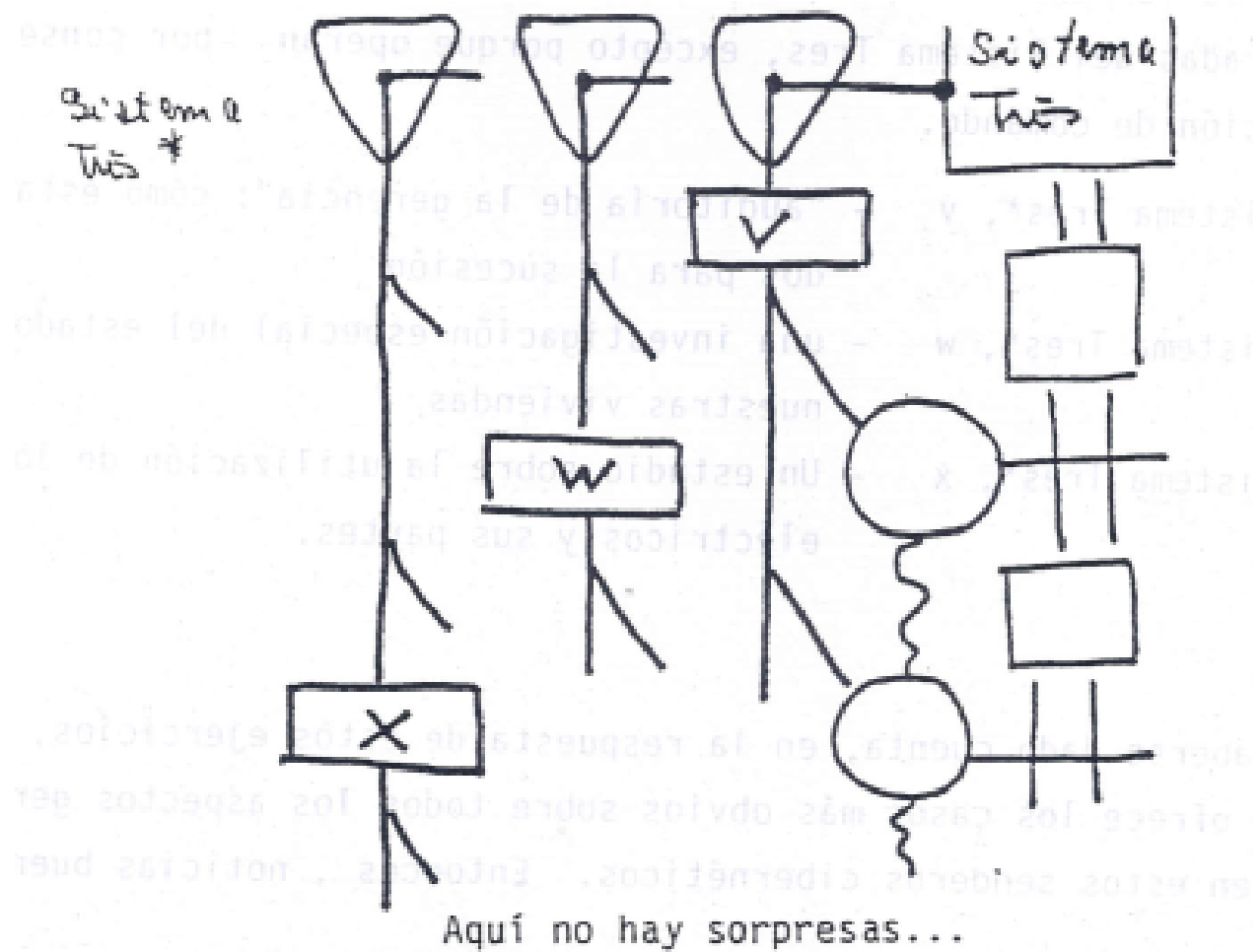


Figure 5: System 3* of the Viable System Model – Faculty Recursion over Departments

Source: Beer (1985).

1. General Description of the System in Focus Within the Viable System Model (VSM), System 3* represents a sporadic, high-variety auditing or inspection mechanism designed to verify the reliability and coherence of information that System 3 receives from System 1 units. Its critical role is to ensure that channels of control are not saturated or distorted due to local filters, errors, or biases (Beer, 1985; Espejo & Reyes, 2011).

In academic performance evaluation, System 3* should function as an institutional verification channel linking the faculty (recursion N) and academic departments (recursion N-1). Its goal is to ensure that department reports, data, and evaluations accurately represent the reality of academic performance.

However, current institutional diagnostics reveal a major weakness: this auditing mechanism is either absent or not systematically implemented. Faculty management receives departmental reports as if they were perfectly consistent, without cross-verification or technical validation. This introduces the risk of systemic dissonance, where formal reports received by System 3 do not always reflect the actual operations performed at System 1 (Beer, 1985; Espejo, 2001).

2. Cybernetic Interpretation of System 3* In the illustration (“No Surprises Here...”), Beer wryly critiques organizations lacking this channel: the control system (S3) blindly trusts the incoming information, assuming no deviations occur. In reality, the absence of independent feedback prevents closure of the control loop, compromising organizational homeostasis (Beer, 1985).

In the Faculty–Department recursion, System 3* should fulfill three essential functions:

- Factual verification of departmental data and reports (e.g., validating evidence, teaching loads, and applied rubrics),
- Calibration of criteria between departments to minimize interpretive divergence,
- Systemic feedback for ongoing platform and instrument improvement (Beer, 1985; Espejo & Harnden, 1989).

3. Proposed Design of System 3* (Avoiding Institutional Resistance) To strengthen the Faculty–Department recursion without introducing a mechanism perceived as punitive control, the design of System 3* should adhere to the principles of intelligent, participatory, and formative auditing, implemented across three complementary levels:

a. Variety Reducers Mechanisms that simplify and standardize audited information, preventing overload:

- Digital consistency matrices that automatically compare commitments, evidence, and outcomes for each academic,
- Cross-validation rubrics utilized by peer reviewers, reducing interpretive ambiguity,
- Automated sampling protocols that randomly select cases or evidence for periodic review, preventing arbitrariness.

b. Variety Amplifiers Mechanisms that increase the verification system’s responsiveness:

- An Academic Data and Monitoring Unit (ADMU): a small, technical non-hierarchical team reporting to the Dean’s Office,
- Lightweight digital audits conducted within the platform using alerts for inconsistencies or duplications,
- Comparative performance dashboards, amplifying process visibility and facilitating the detection of anomalous trends without direct intervention (Beer, 1985; Espejo & Reyes, 2011).

c. Transducers Elements that translate information between operational and strategic domains:

- Automated coherence reports generated by the platform, translating raw workload and performance data into graphical indicators,
- Peer validation surveys that transform qualitative perceptions into consistency metrics,
- Standardized electronic records that convert qualitative reviews into auditable, documented traceability.

Ashby’s Law of Requisite Variety underpins the design, ensuring auditing and transduction mechanisms can process at least the diversity present in the system’s information channels (Ashby, 1956; Beer, 1985).

d. Key Metrics for System 3* Dimension / Key Metric / Description

- *Information Integrity* – % match between department-reported and audit-verified data: Evaluates process reliability.
- *Interdepartmental Consistency* – Standard deviation of average scores across similar departments: Indicates potential biases or divergent criteria.
- *Digital Traceability* – % of agreements and assessments fully registered on the platform: Assesses compliance with institutional workflow.
- *Effective Feedback* – Number of protocol adjustments resulting from 3* review: Indicates systemic learning.

4. Operational Proposal Without Institutional Resistance To ensure the mechanism’s cultural and political viability, it should be positioned as a system for enhancement and institutional learning, not as a hierarchical control:

- *Epistemological positioning*: Name it “Formative Academic Performance Audit” or “Verification and Institutional Learning Module,” emphasizing technical and collaborative functions.
- *Distributed cybernetic design*: Position System 3* not above departments, but as a networked node sharing insights, indicators, and coherence patterns.
- *Automation and transparency*: Conduct verification within the digital platform, with traceability visible to all participants, reducing the perception of surveillance and reinforcing process legitimacy.
- *Feedback loop to System 2*: Channel 3* findings directly inform and improve the protocols and standards managed by System 2, thereby closing the homeostatic stability cycle.

5. Systemic Synthesis System 3* functions as the intelligent verification organ within the academic evaluation model. Its absence results in operational blindness and erodes cross-level trust; its presence fosters transparency, coherence, and organizational learning—without coercion (Beer, 1985; Espejo & Harnden, 1989; Espejo & Reyes, 2011).

Within this model, System 3*:

- Operates from the faculty as a recursive validation and coherence channel, not as an agent of sanctions,
- Uses the digital platform as its main information transducer,
- Closes the cybernetic loop between operational activities (departments) and institutional governance (faculty).

Thus, System 3* does not “police” academics, but rather safeguards the integrity of the evaluative system itself, ensuring institutional results genuinely reflect departmental realities.

3.6 Stage 4. Institutional Learning and Feedback Mechanism

Process: Evidence Collection, Annual Pre-assessment, Biennial Assessment, and Adaptive Learning Based on the OADI Cycle

(See Figure 6: Individual and Organizational Learning Loop Based on the OADI Model)

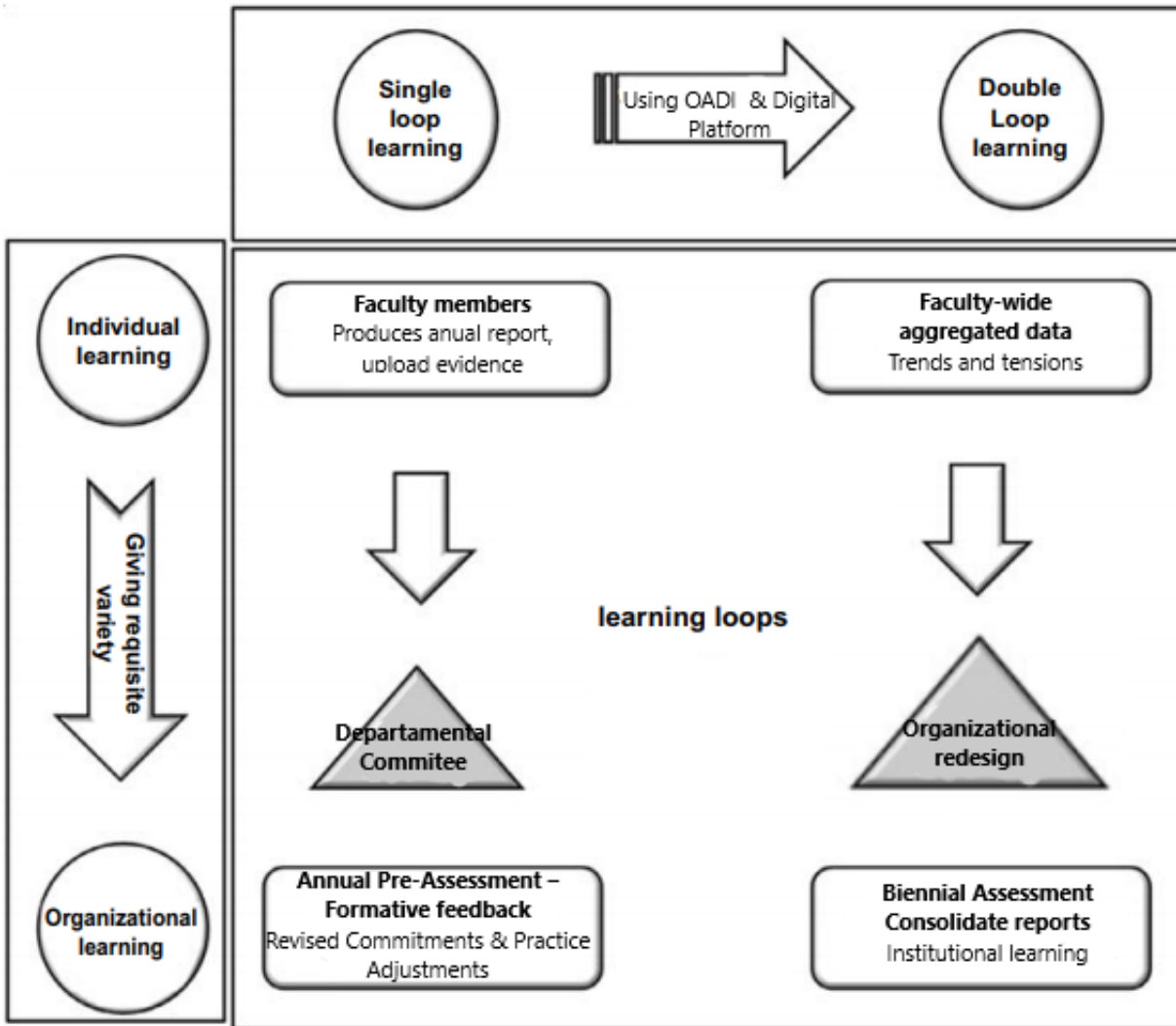


Figure 6: Individual and Organizational Learning Loop Based on the OADI Model

Source: Reyes (2012).

1. General Process Description The Institutional Learning and Feedback Mechanism represents the phase of consolidation, evaluation, and learning within the academic performance process. Its purpose is to ensure that the information generated throughout the evaluative cycle is transformed into actionable knowledge for the continuous improvement of faculty, evaluation committees, and the institution itself (Beer, 1985; Espejo & Reyes, 2011).

The process is temporally structured into two pre-assessment stages and a final biennial assessment, as follows:

First Year:

- Each faculty member prepares and records their annual performance report in the institutional digital platform.
- The report includes commitments from the Performance Agreement, verifiable evidence (course syllabi, publications, research outputs, outreach, and management activities), and a reflective narrative on achievements and challenges.
- The process is governed by the Academic Performance Evaluation Manual, which outlines procedures, criteria, rubrics, and official formats.
- The Departmental Evaluation Committee applies the manual to conduct the first annual pre-assessment, offering formative feedback to the faculty member.

Second Year:

- The process is repeated with an updated evidence portfolio and reported progress.
- The Departmental Committee reviews the academic's progress, assesses improvements, and provides updated feedback.

Biennial Assessment:

- At the end of the second year, the process transitions to the Faculty Committee, which consolidates departmental reports and applies institutional rubrics to issue the final biennial assessment.
- This committee, together with the Dean's Office, ensures cross-departmental coherence, regulatory compliance, and evaluation traceability.

This cyclical structure allows evaluation to function as a progressive, self-regulating learning process, in which feedback from each pre-assessment stage informs the system prior to the final assessment.

2. Institutional Learning Mechanism Based on the OADI Cycle The feedback phase is articulated through the OADI model (Observe–Assess–Design–Implement), translating annual evaluation data into organizational knowledge and improvement. The OADI cycle powers institutional learning by activating observation, analysis, redesign, and implementation processes at both individual and organizational levels (Beer, 1985; Espejo & Reyes, 2011).

a. Observe During each pre-assessment, faculty and committees observe outcomes in relation to declared commitments and submitted evidence.

- The digital platform acts as a transducer, generating comparative reports and dashboards.
- Strengths and performance gaps are identified at both individual and departmental levels.
- Pre-assessment reports provide the basis for longitudinal observation of performance.

b. Assess Observed information is analyzed and evaluated at multiple recursion levels:

- *Individual*: Faculty reflect on practice and adjust strategies (single-loop learning).
- *Departmental*: The Committee reviews criteria, evidence, and rubric application consistency.
- *Faculty*: Aggregated outcomes are examined to detect trends, biases, or structural tensions (e.g., imbalances between teaching and research), enabling double-loop learning.

c. Design Lessons learned translate into redesign:

- Faculty set new commitments and professional development strategies for the next biennial cycle.
- Departmental Committees revise and refine their evaluation and feedback protocols.
- The Faculty, using aggregated data, redefines academic development policies, weighting criteria, career pathways, and support programs.

This adaptive redesign ensures organizational rules and structures evolve with experiential knowledge.

d. Implement Designed changes are implemented in the subsequent cycle, completing the cybernetic feedback loop:

- Improvements are incorporated into the digital platform and institutional manuals.
- Compliance with previous recommendations is monitored.
- Faculty and departments assess the effectiveness of adjustments, restarting the OADI cycle with enhanced learning capacity.

3. Individual and Organizational Learning Loops This mechanism comprises two interconnected loops (see Figure 6):

- *Individual Learning Loop (Pre-assessments)*: Each faculty member receives feedback after each annual pre-assessment, reflects, and adjusts teaching, research, or management practices accordingly. This process nurtures individual self-regulation and continuous improvement.
- *Organizational Learning Loop (Biennial Assessment)*: Committees and the Faculty leverage accumulated information to revise procedures, rubrics, and assessment criteria. The biennial assessment not only synthesizes individual results, but also provides the metasystem with strategic insights on institutional performance, academic productivity, and capacity development.

The Departmental Evaluation Committee serves as a transducer between these levels, converting individual lessons into organizational knowledge, and supplying validated input for faculty-level decision making (Beer, 1985; Espejo & Harnden, 1989).

4. Variety Engineering and Regulation Mechanisms Type / Example / Function

- *Reducers*: Evaluation manual, departmental rubrics, rating protocols, pre-/biennial assessment schedules – Attenuate operational variety, ensuring temporal and methodological consistency.
- *Amplifiers*: Digital platform, comparative reports, collective feedback sessions, improvement workshops – Enhance the system’s capacity to learn from data and adapt.
- *Transducers*: Performance reports, digital portfolios, institutional dashboards, OADI reports – Translate information from individual to organizational levels.

Key Metrics – Alignment between goals and biennial outcomes, improvement compliance rate, cross-departmental coherence, rubric update frequency – measure organizational learning effectiveness and system maturity.

5. Recursive Articulation: Department Faculty The Institutional Learning and Feedback Mechanism operates recursively:

- At the Department level: Annual observation and pre-assessment generate valuable data on individual and collective performance.
- At the Faculty level: The biennial assessment integrates data from all departments, providing a systemic view of academic performance for strategic decision-making.

This recursive flow ensures:

- Each pre-assessment acts as local feedback, fostering self-regulation among faculty and committees.
- The biennial assessment serves as institutional feedback, guiding academic policy and organizational capacity development.

In this way, the system behaves as an intelligent network, capable of self-learning, correcting deviations, and continually improving its evaluation structure (Beer, 1985; Espejo & Reyes, 2011).

6. Synthesis The Institutional Learning and Feedback Mechanism is where performance evaluation evolves into a cybernetic learning process for both individuals and organizations.

Its temporal structure (annual pre-assessment, biennial assessment) enables:

- Ongoing monitoring and support for academic development,
- Adjustment of instruments and criteria before final result consolidation,
- Provision of strategic information for institutional decision-making.

By following the OADI cycle, the system integrates observation, analysis, design, and implementation, closing the organizational learning loop. In this way, the faculty transitions toward becoming a learning organization—capable of self-regulation, anticipatory adaptation, and continuous improvement, grounded in evidence from its own academic performance (Beer, 1985; Espejo & Reyes, 2011).

4 Final Remarks / Discussion

The academic performance evaluation process, interpreted through the lens of organizational cybernetics and the Viable System Model (VSM), reveals a self-regulating structure in which each component—from individual academics to faculty governance bodies—performs a distinct function in maintaining institutional viability (Beer, 1985; Espejo & Reyes, 2011).

The consolidated scheme illustrates how subsystems 1, 2, 3, and 3* are articulated within a recursive circuit of control, feedback, and learning (see figure 7):

- *System 1* (Departments and individual academics) comprises the operational units that generate academic value and the foundational data for performance assessment.
- *System 2* ensures homeostatic coordination, standardizing procedures and dampening oscillation between departments through shared protocols, calendars, and digital platforms.
- *System 3* integrates and regulates overall operations, ensuring consistency between evaluation outcomes and faculty policies.
- *System 3** provides the independent verification mechanism, closing the institutional trust loop through formative audits and integrity reviews (Beer, 1985; Espejo & Harnden, 1989).

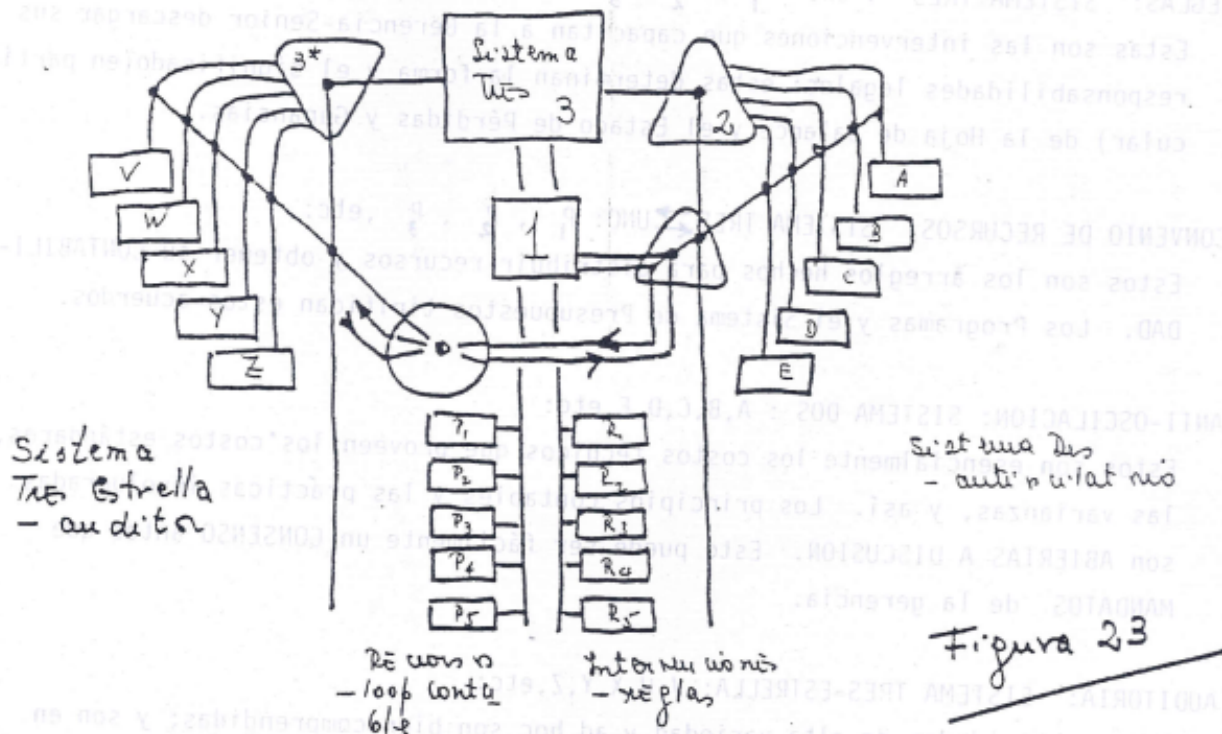


Figure 7: System in Focus – Academic Department Level. Model of System Three with Audit: Interactions, Control Loops, and Feedback

Source: Beer (1985).

Together, these subsystems constitute a loop of control and learning that balances autonomy and cohesion. The annual pre-assessment serves as a mechanism of internal regulation (local feedback), whereas the biennial assessment acts as an instance of global feedback (institutional learning). The inclusion of the Institutional Learning and Feedback Mechanism, grounded in the OADI cycle (Observe–Assess–Design–Implement), enables the faculty not only to evaluate performance but also to learn from itself, continuously adjusting criteria, instruments, and policies based on evidence generated by its own system (Espejo & Reyes, 2011).

This model, by integrating principles of variety engineering (reducers, amplifiers, and transducers) and organizational double-loop learning, transforms evaluation into an essential cybernetic function:

- Complexity is reduced through shared standards.
- Responsiveness is amplified via digitalization and interdepartmental coordination.
- Qualitative information is transduced into strategic knowledge for academic planning.

From this perspective, performance evaluation ceases to be merely an administrative procedure and becomes a system of academic governance that embeds control, transparency, and learning. The faculty thus achieves a dynamic balance between stability and change: remaining faithful to its institutional identity while evolving through each evaluation cycle.

In summary, the consolidated model is not only a tool for measuring performance, but also a framework for self-reflection—turning evaluation into a recursive practice of self-regulation and continuous improvement. It is, in essence, a truly viable system for institutional learning (Beer, 1985; Espejo & Reyes, 2011).

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